

The Centered Fixed-Trace GUE Large-Deviation Principle

A full proof of the conjecture in arXiv:2508.13698

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Status: candidate new full solution. The conjectured large-deviation principle is proved at speed n^2 by a direct microcanonical logarithmic-gas argument and contraction through the centering map. Independent expert review is recommended, especially for the standard quantile-patch lower-bound lemma recorded in full below.

1 Source conjecture

Chaintron and Lacker ask on PDF page 35 of [1] for the large deviations of the centered empirical eigenvalue law of the fixed-trace Gaussian Unitary Ensemble (GUE). Their exact setup and proposed rate are reproduced below.

Let $\mathcal{H}_n$ be the real Hilbert space of $n \times n$ Hermitian matrices with Hilbert–Schmidt norm. Let M_n be uniform on

$$\{H \in \mathcal{H}_n : \text{tr}(H^2) = n^2\}.$$

Write $x_1^{(n)}, \dots, x_n^{(n)}$ for the eigenvalues of $M_n/\sqrt{n}$ and put

$$L_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i^{(n)}}, \quad \mathbb{C}\mu = (T_{-m_1(\mu)})_{\#}\mu, \quad T_a(x) = x + a.$$

Thus $\mathbb{C}L_n$ is precisely the centered empirical measure displayed in the source.

For a probability measure μ , define the signed mean and quadratic moment

$$m_1(\mu) = \int x \mu(\mathrm{d}x), \quad m_2(\mu) = \int x^2 \mu(\mathrm{d}x), \quad \mathcal{E}_{\log}(\mu) = - \iint \log |x - y| \mu(\mathrm{d}x) \mu(\mathrm{d}y),$$

with $\mathcal{E}_{\log} = +\infty$ when the integral diverges positively. The source uses

$$I_{\log}(\mu) = \mathcal{E}_{\log}(\mu) + \frac{1}{2}m_2(\mu) - \frac{3}{4}.$$

2 Result

We use the standard convention that an LDP with speed n^2 and rate J means

$$-\inf_G J \leq \liminf_{n \rightarrow \infty} \frac{1}{n^2} \log \mathbb{P}(X_n \in G)$$

for every open G , and the reverse upper bound for every closed set.

Proof. Theorem 4.3 (with $\theta = 1$) shows that the functional $I_{\log}(\cdot)$ is 1-convex in $\mathcal{S}_{0,1}$. Because σ is the unique minimizer of $I_{\log}$ on $\mathcal{P}_2(\mathbb{R})$, it is also the unique minimizer on $\mathcal{S}_{0,1}$. Using Proposition 2.2 along with $I_{\log}(\sigma) = 0$ proves the claim. $\square$

In the spirit of Section 5.1, it is natural to expect the inequality (5.4) to relate to the aforementioned large deviations principle for Wigner matrices. The Gaussian Unitary Ensemble (GUE) is the probability measure on the space $\mathcal{H}_n$ of $n \times n$ Hermitian matrices whose density with respect to Lebesgue measure is proportional to $H \mapsto \exp(-\text{tr}(H^2)/2)$. The associated *fixed trace ensemble*, formally obtained by conditioning on $\text{tr}(H^2) = n^2$, has the same expression for its density relative to the surface measure on $\{H \in \mathcal{H}_n : \text{tr}(H^2) = n^2\}$. Let M_n be a random matrix drawn from this fixed trace ensemble, and let $\lambda_1^n < \dots < \lambda_n^n$ denote the eigenvalues of $M_n/\sqrt{n}$. Consider the centered empirical measure

$$CL_n = \frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i^n - \frac{1}{n} \sum_{j=1}^n \lambda_j^n}.$$

We conjecture that CL_n satisfies a large deviation principle in $(\mathcal{P}_1(\mathbb{R}), \mathcal{W}_1)$ with good rate function

$$\mu \mapsto \begin{cases} I_{\log}(\mu) + \frac{1}{2}(1 - m_2(\mu)) & \text{if } m_1(\mu) = 0, \ m_2(\mu) \leq 1, \\ \infty & \text{otherwise,} \end{cases}$$

where again $m_p(\mu) = \int_{\mathbb{R}} x^p \mu(dx)$ for $p \geq 1$. It is not difficult to show (similarly to Proposition 5.1) that this conjectural rate function is the weakly lower-semicontinuous envelope of

$$\mu \mapsto \begin{cases} I_{\log}(\mu) & \text{if } \mu \in \mathcal{S}_{0,1} \\ \infty & \text{otherwise.} \end{cases}$$

With this large deviation principle in hand, it would follow as in Proposition 5.3 that

$$\limsup_{m \rightarrow \infty} \frac{1}{n} \log \mathbb{P}(\mathcal{W}_1(CL_n, \sigma) \geq r) \leq -2 \arcsin^2(r/2),$$

for $0 < r \leq 2$, with the right-hand side replaced by $-\infty$ for $r > 2$. A simpler consequence in terms

Figure 1: The fixed-trace GUE conjecture in [1, p. 35].

Theorem 1 (Fixed-trace GUE conjecture). *The sequence (CL_n) satisfies an LDP in $(\mathcal{P}_1(\mathbb{R}), \mathcal{W}_1)$ with speed n^2 and good rate function*

$$J(\mu) = \begin{cases} I_{\log}(\mu) + \frac{1}{2}(1 - m_2(\mu)), & m_1(\mu) = 0, \ m_2(\mu) \leq 1, \\ +\infty, & \text{otherwise.} \end{cases}$$

Equivalently, on its effective domain,

$$J(\mu) = \mathcal{E}_{\log}(\mu) - \frac{1}{4}.$$

This is exactly the rate conjectured in [1, p. 35].

The proof has two independent pieces. First, diagonalization turns the fixed-trace matrix ensemble into a logarithmic gas on a Euclidean sphere. Second, centering is a continuous map in the topology requested by the source, and its contraction fiber can be computed explicitly.

3 The spherical logarithmic gas

Put

$$\Sigma_n = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i^2 = n \right\}$$

and let S_n denote its surface measure. Weyl's integration formula, followed by the coarea formula and the change from the eigenvalues of M_n to those of $M_n/\sqrt{n}$, gives the unordered joint eigenvalue law

$$Q_n(\mathrm{d}x) = \frac{1}{Z_n} \prod_{1 \leq i < j \leq n} |x_i - x_j|^2 S_n(\mathrm{d}x), \quad x \in \Sigma_n. \quad (1)$$

All dimension-dependent constants are absorbed into Z_n . Ordering the eigenvalues only contributes a factor $n!$, whose logarithm is $o(n^2)$.

Proposition 1 (Microcanonical log-gas LDP). *Under (1), the empirical measure $L_n = n^{-1} \sum_i \delta_{x_i}$ satisfies an LDP in $(\mathcal{P}_1(\mathbb{R}), \mathcal{W}_1)$ with speed n^2 and good rate*

$$K(\nu) = \begin{cases} \mathcal{E}_{\log}(\nu) - \frac{1}{4}, & m_2(\nu) \leq 1, \\ +\infty, & m_2(\nu) > 1. \end{cases} \quad (2)$$

Moreover,

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \log Z_n = -\frac{1}{4}. \quad (3)$$

The domain $m_2 \leq 1$, rather than $m_2 = 1$, is essential. At every finite n one has $m_2(L_n) = 1$, but $\mathcal{W}_1$ convergence can lose quadratic moment. The lost moment is carried by one eigenvalue of order $\sqrt{n}$.

3.1 Compactness and lower semicontinuity

Let

$$\mathcal{D} = \{\nu \in \mathcal{P}_1(\mathbb{R}) : m_2(\nu) \leq 1\}.$$

This set is $\mathcal{W}_1$ -compact. Indeed, the quadratic bound gives tightness and

$$\sup_{\nu \in \mathcal{D}} \int_{|x| > R} |x| \nu(\mathrm{d}x) \leq R^{-1},$$

which gives uniform integrability of first moments. It is also closed by Fatou's lemma.

The logarithmic energy is lower semicontinuous on $\mathcal{D}$. A convenient way to see the point at infinity is the identity

$$h(x, y) = -\log|x - y| + \frac{1}{2} \log(1 + x^2) + \frac{1}{2} \log(1 + y^2) \geq 0, \quad (4)$$

$$\mathcal{E}_{\log}(\nu) = \iint h(x, y) \nu(\mathrm{d}x) \nu(\mathrm{d}y) - \int \log(1 + x^2) \nu(\mathrm{d}x). \quad (5)$$

The inequality follows from $|x - y|^2 \leq (1 + x^2)(1 + y^2)$. The first term is lower semicontinuous by Portmanteau, while the second is continuous under $\mathcal{W}_1$ convergence because $\log(1 + x^2) = o(|x|)$. Hence K is lower semicontinuous on the compact set $\mathcal{D}$, and is a good rate once the bounds are proved.

Finally, the classical semicircle variational principle says

$$\inf_{\nu \in \mathcal{D}} \mathcal{E}_{\log}(\nu) = \frac{1}{4}, \quad (6)$$

with equality at the centered variance-one semicircle law σ . For completeness, this follows immediately from the standard inequality $I_{\log}(\nu) \geq 0$: if $m_2(\nu) \leq 1$, then

$$\mathcal{E}_{\log}(\nu) \geq \frac{3}{4} - \frac{1}{2}m_2(\nu) \geq \frac{1}{4},$$

and the source records $m_2(\sigma) = 1$ and $\mathcal{E}_{\log}(\sigma) = 1/4$.

3.2 Upper bound

For $\varepsilon > 0$, set

$$\Phi_{\varepsilon}(\nu) = \iint \log(|x - y| \vee \varepsilon) \nu(dx) \nu(dy).$$

On $\mathcal{D}$ this functional is continuous in $\mathcal{W}_1$; its logarithmic growth at infinity is uniformly integrable under the quadratic bound. For $x \in \Sigma_n$,

$$2 \sum_{i < j} \log |x_i - x_j| \leq n^2 \Phi_{\varepsilon}(L_n) - n \log \varepsilon. \quad (7)$$

Also $\log S_n(\Sigma_n) = O(n)$ by the exact sphere-area formula. Therefore, for a closed $F \subset \mathcal{P}_1(\mathbb{R})$,

$$\limsup_{n \rightarrow \infty} \frac{1}{n^2} \log \int_{\{L_n \in F\}} \prod_{i < j} |x_i - x_j|^2 S_n(dx) \leq \sup_{\nu \in F \cap \mathcal{D}} \Phi_{\varepsilon}(\nu). \quad (8)$$

The decreasing limit of Φ_{ε} as $\varepsilon \downarrow 0$ is $-\mathcal{E}_{\log}$. Compactness of $F \cap \mathcal{D}$ and the usual decreasing-maxima lemma for upper semicontinuous functions give

$$\limsup_{n \rightarrow \infty} \frac{1}{n^2} \log \int_{\{L_n \in F\}} \prod_{i < j} |x_i - x_j|^2 S_n(dx) \leq \sup_{\nu \in F \cap \mathcal{D}} [-\mathcal{E}_{\log}(\nu)]. \quad (9)$$

Taking $F = \mathcal{P}_1(\mathbb{R})$ and using (6) yields

$$\limsup n^{-2} \log Z_n \leq -\frac{1}{4}.$$

3.3 Lower bound and the escaped eigenvalue

We give the construction because it is the only noncanonical feature of the argument. First suppose that ν has compact support, a bounded density, finite logarithmic energy, and $m_2(\nu) < 1$. Choose $n - 1$ separated quantiles $y_1 < \dots < y_{n-1}$ of ν , and set

$$b_n = \left(n - \sum_{i=1}^{n-1} y_i^2 \right)^{1/2}. \quad (10)$$

Then

$$b_n = \sqrt{n(1 - m_2(\nu))} + o(\sqrt{n}).$$

In particular, the empirical law of $(y_1, \dots, y_{n-1}, b_n)$ converges to ν in $\mathcal{W}_1$: the last point has mass $1/n$ and contributes only $O(n^{-1/2})$ to the first moment.

Take disjoint intervals of radius c/n around the bulk quantiles, with c fixed sufficiently small, and parametrize the positive cap of Σ_n by

$$x_n = \left(n - \sum_{i=1}^{n-1} x_i^2 \right)^{1/2}. \quad (11)$$

The surface Jacobian is

$$\sqrt{1 + |\nabla x_n|^2} = \frac{\sqrt{n}}{x_n},$$

which stays bounded above and below on this patch. Its logarithmic surface volume is $-O(n \log n) = o(n^2)$.

If the density of ν is bounded by A , then $y_j - y_i \geq (j - i)/(An)$. Shrinking c if necessary shows uniformly on the patch that

$$\sum_{i < j < n} \log |x_i - x_j| \geq \sum_{i < j < n} \log |y_i - y_j| - O(n \log n).$$

The quantile Riemann-sum lemma for the locally integrable logarithmic kernel gives

$$\lim_{n \rightarrow \infty} \frac{2}{n^2} \sum_{i < j < n} \log |y_i - y_j| = \iint \log |x - y| \nu(dx) \nu(dy) = -\mathcal{E}_{\log}(\nu). \quad (12)$$

All $n - 1$ interactions with $x_n \asymp \sqrt{n}$ total only $O(n \log n)$. Thus every $\mathcal{W}_1$ -neighborhood G of ν satisfies

$$\liminf_{n \rightarrow \infty} \frac{1}{n^2} \log \int_{\{L_n \in G\}} \prod_{i < j} |x_i - x_j|^2 S_n(dx) \geq -\mathcal{E}_{\log}(\nu). \quad (13)$$

The regularity assumptions cost nothing. If $\mathcal{E}_{\log}(\nu) < \infty$ and $m_2(\nu) < 1$, first truncate, then convolve with a smooth compactly supported density, and if needed apply a dilation arbitrarily close to one. The resulting bounded-density compactly supported measures converge in $\mathcal{W}_1$, retain second moment strictly below one, and their logarithmic energies converge to $\mathcal{E}_{\log}(\nu)$. This is the standard approximation lemma for logarithmic energy; it follows directly by first truncating the nonnegative kernel h in (4) and then using monotone convergence. If $m_2(\nu) = 1$, apply the preceding bound to $(D_s)_\# \nu$, where $D_s(x) = sx$ and $s < 1$, and let $s \uparrow 1$. Indeed,

$$\mathcal{E}_{\log}((D_s)_\# \nu) = \mathcal{E}_{\log}(\nu) - \log s. \quad (14)$$

The lower bound is vacuous when $\mathcal{E}_{\log}(\nu) = +\infty$.

Applying (13) to $(D_s)_\# \sigma$ and letting $s \uparrow 1$ yields

$$\liminf n^{-2} \log Z_n \geq -\frac{1}{4}.$$

This proves (3). Subtracting the normalizer from (9) and (13) proves the LDP of Proposition 1. Since every L_n lies in the fixed compact set $\mathcal{D}$, exponential tightness is automatic.

4 Centering and completion of the proof

Lemma 1 (Centering is continuous). *The map $\mathbb{C} : (\mathcal{P}_1(\mathbb{R}), \mathcal{W}_1) \rightarrow (\mathcal{P}_1(\mathbb{R}), \mathcal{W}_1)$ is 2-Lipschitz.*

Proof. For any coupling of μ and ν , $|m_1(\mu) - m_1(\nu)| \leq \mathcal{W}_1(\mu, \nu)$. Translating the same coupling gives

$$\mathcal{W}_1(\mathbb{C}\mu, \mathbb{C}\nu) \leq \mathcal{W}_1(\mu, \nu) + |m_1(\mu) - m_1(\nu)| \leq 2\mathcal{W}_1(\mu, \nu).$$

□

By Proposition 1, Lemma 1, and the contraction principle, CL_n has good rate

$$J(\mu) = \inf\{K(\nu) : C\nu = \mu\}. \quad (15)$$

If $m_1(\mu) \neq 0$, the fiber is empty. If $m_1(\mu) = 0$, its members are exactly the translates $(T_a)_\# \mu$, $a \in \mathbb{R}$. Translation invariance of the logarithmic energy and centering of μ give

$$\mathcal{E}_{\log}((T_a)_\# \mu) = \mathcal{E}_{\log}(\mu), \quad m_2((T_a)_\# \mu) = m_2(\mu) + a^2. \quad (16)$$

The fiber contains an admissible member of $\mathcal{D}$ exactly when $m_2(\mu) \leq 1$. In this case every admissible member has the same K -value, so (15) becomes

$$J(\mu) = \mathcal{E}_{\log}(\mu) - \frac{1}{4} = I_{\log}(\mu) + \frac{1}{2}(1 - m_2(\mu)).$$

This proves Theorem 1.

5 Checks, consequences, and scope

1. **Speed.** The Vandermonde has $\binom{n}{2}$ interactions, so the speed is n^2 . The final display on source page 35 has $\limsup_{m \rightarrow \infty} n^{-1} \log$; this appears to be an m/n and speed typographical error. The LDP proved here gives $n^{-2} \log$.
2. **Why no quadratic penalty remains.** On the fixed sphere, the quadratic external-field term is constant. Algebraically, $I_{\log} + (1 - m_2)/2 = \mathcal{E}_{\log} - 1/4$. Geometrically, missing quadratic moment can escape through one coordinate without an n^2 -scale interaction cost.
3. **Centering does not hide a discontinuity.** The first moment is continuous in $\mathcal{W}_1$, and Lemma 1 gives a quantitative continuity estimate. No $\mathcal{W}_2$ contraction is invoked.
4. **Concentration consequence.** Combining this LDP with the lower-semicontinuous extension of the strengthened Biane–Voiculescu inequality in [1] gives the tail bound proposed there, with corrected speed n^2 .
5. **Scope.** The proof is for the complex Hermitian ($\beta = 2$) ensemble stated in the source. The same argument formally replaces $\mathcal{E}_{\log}$ by $\frac{\beta}{2} \mathcal{E}_{\log}$ for fixed-trace β -Hermite laws, but no such extension is claimed here.

6 Novelty search and review recommendation

A bounded search on August 11, 2026 checked the exact conjecture, the arXiv identifier, “fixed trace GUE empirical measure large deviations”, “fixed trace beta-Hermite LDP”, and the lost-second-moment rate. It found the classical canonical Wigner empirical-measure LDP [2], work on fixed-trace beta-Hermite limiting density and edge behavior (arXiv:0905.4255), and large deviations for traces/moments (arXiv:1605.03894), but no source stating the $\mathcal{W}_1$ microcanonical LDP (2) or its centered contraction. The result is therefore labeled a candidate new full proof rather than a literature answer.

The recommended expert audit should focus on the quantile-patch estimate (12), the regularization step following (13), and whether a later fixed-trace ensemble paper contains the same theorem under different terminology. The contraction calculation itself is exact and elementary.

References

- [1] L.-P. Chaintron and D. Lackner, *Geodesic convexity and strengthened functional inequalities in submanifolds of Wasserstein space*, arXiv:2508.13698 (2025).
- [2] G. Ben Arous and A. Guionnet, *Large deviations for Wigner's law and Voiculescu's non-commutative entropy*, Probab. Theory Related Fields **108** (1997), 517–542.
- [3] P. Biane and D. Voiculescu, *A free probability analogue of the Wasserstein metric on the trace-state space*, Geom. Funct. Anal. **11** (2001), 1125–1138.